Numerical Analysis

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1 Trapezium Rule

Q3.1 Defined a function with the integrand:

$$f(x) = \frac{x^4(1-x)^4}{1+x^2}$$

Q3.2 Using the `trap0` function with 100 trapezia, the value of the integral was found to be:

0.0012644892673714441

Q3.3 To obtain the value accurate to 6 significant figures, lower numbers of trapezia were used until the first 6 digits matched the 100-trapezia value. It was found that 18 trapezia gives:

0.0012644899090426126

2 Specified Accuracy

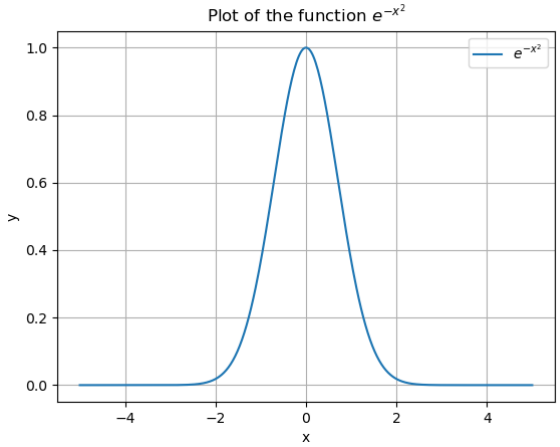


Figure 1: Plot of e^{-x^2} to determine suitable integration limits.

Q3.4 Using the integrand e^{-x^2} along with the `trap1` function (specified accuracy), we can compute the integral. Limits of infinity are not practical in code, so from Fig. 1 we observe that the function effectively vanishes after $|x| \approx 4$. Using limits $x \in [-4, 4]$, the integral is:

$$\int_{-4}^{4} e^{-x^2} dx \approx 1.772454$$

To verify, limits $[-10, 10]$ give:

$$\int_{-10}^{10} e^{-x^2} dx \approx 1.772454$$

There is almost no difference suggesting the limits $[-4, 4]$ are good enough.

3 Pendulum Problem

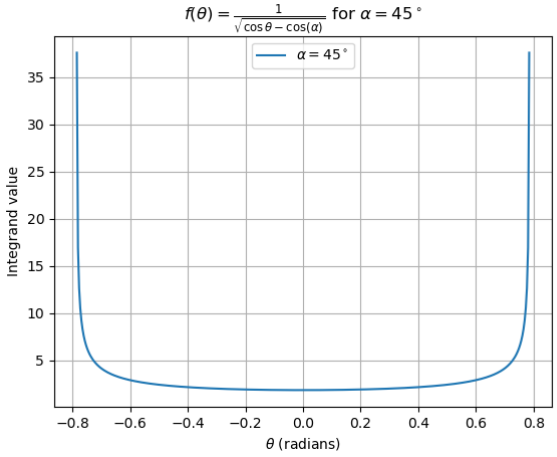


Figure 2: Plot of the pendulum integrand for different amplitudes.

Q3.6 Using the `trap1` function with a range of α values, a set of T/T_0 ratios were found, as shown below:

α (degrees)	T/T_0
0	1.000000
15	1.004301
30	1.017409
45	1.039973
60	1.073182
75	1.118959
90	1.180341

Table 1: Pendulum period ratios for different amplitudes.

Q3.7 To achieve higher accuracy for $\alpha = 90°$, the `trap1` function was rerun with a smaller delta to reach an accuracy of 10^{-12} instead of 10^{-6}. The resulting value is:

$$T/T_0 = 1.1803405990160964$$

4 Monte Carlo Simulation

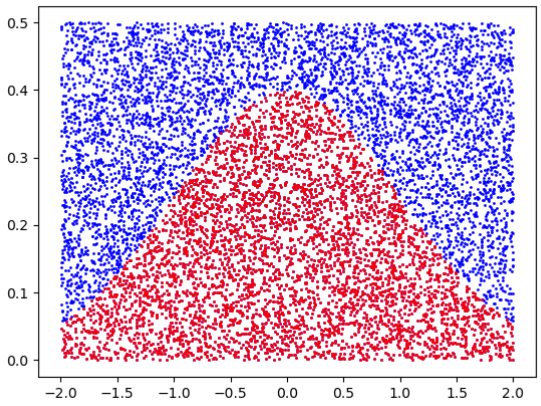


Figure 3: Representation of the Monte Carlo simulation.

Q3.8 To perform a Monte Carlo simulation, a set of x and y points were generated using the `random.uniform` function for 10,000 points within ± 2 standard deviations and up to a probability of 0.5.

The simulated data were then compared to a Gaussian function generated using `norm.pdf(random_x, 0, 1)`, which produces a normal distribution with mean 0 and standard deviation 1 at each x point. This allows each y point to be tested as either under or above the normal distribution.

The fraction of points under the curve, multiplied by the total area, gives the estimated area under the curve. Fig. 3 illustrates the points under the distribution in red and those above in blue.

The probability within ± 2 standard deviations found using this method is approximately:

$$0.9704$$